

**Sigmoid Activation Derivative:**

Let, sigmoid of z be,  $s(z) = \frac{1}{1 + e^{-x}} = (1 + e^{-x})^{-1}$

the derivative of sigmoid,  $s'(z) = - (1 + e^{-x})^{-2} \times (-e^{-x})$

$$\begin{aligned} &= \frac{e^{-x}}{(1 + e^{-x})^2} \\ &= \frac{1}{(1 + e^{-x})} \times \frac{e^{-x}}{(1 + e^{-x})} \\ &= s(z) \times \frac{(1 + e^{-x}) - 1}{(1 + e^{-x})} \\ &= s(z) \times \left( 1 - \frac{1}{(1 + e^{-x})} \right) \\ &= s(z)(1 - s(z)) \end{aligned}$$